VIT Vidyalankar Institute of Technology
Accredited A+ by NAAC

Version 09/27-09

Consolidated Academic Administration Plan for the Course

Machine Learning Sem.V – Program INFT 2026-27 Odd Semester

Faculty – Prof. Dr. Suja Jayachandran Cluster Mentor) & Prof. Kanchan Dhuri

The academic resources available in VIT –

VMIS (ERP)	V-Refer and V-Live	VIT Library	VAC & MOOC Courses
Institute & Department Vision and Mission	Former IA question papers and solutions (prepared by faculty)	Former IA question papers solutions - hardcopy	Value Added Courses (VAC) are conducted throughout the semester & in the semester break - Enrol for the VACs
Program Educational Objectives (PEO)	MU end semester examination question papers and solutions (prepared by faculty)	MU end semester exam question paper & solutions - by faculty, hardcopy	
Program Specific Outcome (PSO)	Class notes and Digital Content for the subject (scanned / typed by faculty)	All text books, reference books, e -books mentioned in the syllabus & AAP	Online courses from NPTEL, Coursera etc. are pursued throughout the semester - Register for the course & get certified
Program Outcome (PO)	Comprehensive question bank, EQ, GQ, PPT, Class Test papers	Technical journals and magazines for reference	
Departmental Knowledge Map	Academic Administration Plan & Beyond Syllabus Activity report	VIT library has many resources e:g :- IEEE, Nimbus, xplore, EBSCO etc.	Watch former lectures captured in LMS at VIT

1.a Course Objectives (Write in detail – as per NBA guidelines)

Cognitive	What do you want students to know?	Basic concepts of Machine Learning
Affective	What do you want students to think / care about?	Identifying machine learning techniques suitable for a given problem
Behavioural	What do you want students to be able to do?	Solve the problems using various machine learning techniques, apply dimensionality reduction techniques and design applications using machine learning frameworks.

Advice to Students:

Attend every class!!! Missing even one class can have a substantial effect on your ability to understand the course. Be prepared to think and concentrate, in the class and outside. I will try to make the class very interactive. Participate in the class discussions. Ask questions when you don't understand something. Keep up with the class readings. Start assignments and homework early. Meet me in office hour to discuss ideas, solutions or to check if, what you understand is correct.

The v-Refer Link

Example:http://vidyalankarlive.com/vrefer/index.php/apps/files/?dir=/vRefer/FE/SEM%20I/202122/Engineering%20Mechanics/GM&fileid=586945 /

Creation of microsite or teams link – (Division wise)

General | INFT SEM 5 Machine Learning Theory Div A

General | INFT SEM 5 Machine Learning Theory Div B

General | INFT SEM 5 Machine Learning Theory Div C

Collaboration Policy:

We encourage discussion between students regarding the course material. However, no discussion of any sort is allowed with anyone on the assignment and homework for the class. If you find solution to some problems in a book or on the internet, you may use their idea for the solution; provided you acknowledge the source (name and page in the book or the website, if the idea is found on the internet). Even though you are allowed to use ideas from another source, you must write the solution in your own words. If you are unsure whether or not certain kinds of collaboration is possible, please ask the teacher.

1.b Course Outcome (CO) Statements and Module-Wise Mapping (follow NBA guideline)

CO No.	Statements	Related Module/s
CO1	Recall fundamental concepts, terminologies, and categories of machine learning including supervised, unsupervised, and reinforcement learning techniques.	1
CO2	Explain the working principles of dimensionality reduction, regression, classification, clustering, and neural network models used in machine learning.	1, 2, 3, 4
CO3	Apply suitable machine learning algorithms such as regression, SVM, Random Forest, clustering, and perceptron models to solve basic data-driven problems.	2, 4, 5
CO4	Analyze the performance of machine learning models using evaluation metrics, bias-variance analysis, regularization methods, and ensemble techniques.	3
CO5	Evaluate and optimize machine learning solutions using hyperparameter tuning, model selection strategies, and optimization approaches for real-world scenarios.	3, 6
CO6	Design basic machine learning workflows integrating preprocessing, model building, evaluation, and optimization techniques for practical applications.	1 to 6

1.c Course Outcome (CO) Statements mapping with Learning Outcome (LO)

	Mapped to Learning Outcomes
CO1	LO 1.1, LO 5.1
CO2	LO 1.2, LO 1.3, LO 2.1, LO 4.1
CO3	LO 2.2, LO 2.3, LO 5.2, LO 3.3
CO4	LO 3.1, LO 3.2, LO 4.2, LO 4.3, LO 6.1, LO 6.2
CO5	LO 5.3, LO 6.3
CO6	LO 1.3, LO 2.2, LO 2.3, LO 3.1, LO 4.2, LO 5.2, LO 6.3

1.d Mapping of COs with POs (As per new detailed Syllabus)

(List of POs is available in V-refer)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
	Knowledge A	Analysis	Design	Investigation	Modern Tools	Society	Environment &sustainability	Ethics T	Teamwork C	Communication	Project Mgt	Lifelong learning
CO 1	2/18											
CO 2	3/18	1/18										
CO 3	3/18	1/18										
CO 4	1/18	3/18		2/18								
CO 5		1/18	1/18									
CO 6		1/18	1/18									

1.e Mapping of COs with PSOs (As per new detailed Syllabus)

	PSO 1	PSO 2	PSO 3
CO 1	2/18		
CO 2	4/18		
CO 3	4/18		
CO 4	4/18	2/18	
CO 5	2/18		
CO6	2/18		

1.f Teaching and Examination Scheme (As specified by the autonomous syllabus) for the Course

Verticals	BSC/ESC	Program Courses	Multidisciplinary Courses	Skill Courses	HSSM	Experiential Learning	Liberal Learning
Tick suitable category		√					

Subject Code	Subject Name	**Teaching Scheme**			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/Practical	Tutorial	Total

PCIT15T	Machine Learning	2	-	-	2	-	-	2
PCIT15P	Machine Learning Lab	-	2	-	-	-	-	1

Subject Code	Subject Name				**Examination Scheme**				
		Theory			**Total (Theory)**	**Practical**			**Total (Practical)**
		ISA	MSE	ESE		ISA	ESE	ORAL	
PCIT15T	Machine Learning	15	20	40	75				
PCIT15P	Machine Learning Lab					25	-	25	50

Subject Code	Subject Name	**MSE-1***		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PCIT15T	Machine Learning	Q1.a	M1-10M	BL 3
		Q1.b	M1-10M	BL2, 3
		Q2.a	M1-10M	BL2, 3
		Q2.b	M2-10M	BL2, 3

Subject Code	Subject Name	**MSE-2***		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PCIT15T	Machine Learning	Q1.a	M3-5M	BL2, 3
		Q1.b	M4-5M	BL2, 3
		Q1.c	M5-5M	BL2, 3
		Q2.a	M3-10M	BL2, 3
		Q2.b	M4-10M	BL2, 3

Subject Code	Subject Name	**ESE#**		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PCIT15T	Machine Learning	Q1.a	M1-10M	BL2, 3
		Q1.b	M2-10M	BL2, 3
		Q2.a	M3-10M	BL2, 3
		Q2.b	M4-10M	BL2, 3
		Q3.a	M5-10M	BL2, 3
		Q3.b	M6-10M	BL2, 3
		Q4.a	M3-10M	BL2, 3
		Q4. b	M4-10M	BL2, 3

* **Recommended distribution: -** 30 Marks from Assignments, 40 marks based on assignments with slightly enhance difficulty /complex, 30 marks from thought provoking

Recommended distribution: - 30 Marks from Assignments, 40 marks based on assignments/MSE with slightly enhance difficulty /complex, 30 marks from thought provoking

1.g Faculty-Wise Distribution of all Lecture-Practical-Tutorial Hours for the Course

Divisions	Lecture (Hrs.)	Practical (Hrs.)				Tutorial (Hrs.)			
		Batch 1	Batch 2	Batch 3	Batch 4	Batch 1	Batch 2	Batch 3	Batch 4
A	(KGD) 2	2	2	2	2	-	-	-	-
B	(KGD) 2	2	2	2	2	-	-	-	-
C	(KGD) 2	2	2	2	2	-	-	-	-

1.h Office Hours (Faculty will be available in office in this duration for solving students' query)

Division	Day	Time (at least 1 Hr. / Division)	Venue (Office Room No.)
A	Tuesday	4.00PM – 5.00PM	E105 INFT Staff room
B	Thursday	4.00PM – 5.00PM	E105 INFT Staff room
C	Monday	4.00PM – 5.00PM	E105 INFT Staff room

Note: Office hours should be after academic hours of students.

2.a Syllabus: Module Wise Teaching Hours and % Weightage in autonomous syllabus Question Paper

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
1	Introduction to Machine Learning Introduction to Machine Learning Techniques: Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data formats, Applications, Feature Selection and Filtering, Dimensionality Reduction Techniques, Principal Component Analysis, Linear Discriminant Analysis, Singular Valued Decomposition.	6	20	30	10	PI 1.4.1, PI 1.4.2, PI 2.2.5
Learning Outcome-1.1	Define key machine learning paradigms: supervised, unsupervised, and reinforcement learning. (PI 1.4.1) (CO1)					
Learning Outcome-1.2	Explain the importance of data representation, feature selection, and dimensionality reduction. (PI 1.4.2) (CO2)					
Learning Outcome-1.3	Describe the applications and scope of machine learning across domains. (PI 2.2.5) (CO2)					
2	Supervised Learning-I Regression: Linear regression models, Nonlinear regression (only introduction), SVM classifier Support Vector Machine	6	20	25	10	PI 1.4.2 PI 1.4.3 PI 2.4.3

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
	classification algorithm, hyper plane, optimal separating hyperplanes, kernel functions, kernel selection, applications. Introduction to random forest, growing of random forest, random feature selection					
Learning Outcome-2.1	Explain the working of linear and nonlinear regression models. (PI 1.4.2) (CO2)					
Learning Outcome-2.2	Apply SVM for binary classification problems using kernel functions. (PI 1.4.3) (CO3)					
Learning Outcome-2.3	Demonstrate the use of Random Forests in classification tasks. (PI 2.4.3) (CO3)					
3	Supervised Learning-II Evaluation of classifiers: Accuracy, Precision, Recall, F1 score, TPR, TNR, Confusion matrix, ROC, Overfitting, Underfitting, Variance, Bias, Bias-Variance Trade-off, Concepts of regularization (L1 and L2) and generalization, Hyperparameter Tuning, Ensemble Learning: Basic concept, Stacking, Bagging, Boosting, Random Forest, AdaBoost, GBM, XG Boost	6	20	20	20	PI 4.3.3 PI 2.4.4 PI 1.4.3
Learning Outcome-3.1	Compute and interpret evaluation metrics such as accuracy, precision, recall, and F1 score. (PI 4.3.3) (CO4)					
Learning Outcome-3.2	Analyze the effects of overfitting, underfitting, bias, and variance on model performance. (PI 2.4.4) (CO4)					
Learning Outcome-3.3	Apply ensemble learning methods like Bagging, Boosting, and Stacking. (PI 1.4.3) (CO3)					
4	Unsupervised Learning Types of Clustering algorithms, Graph Based Clustering: Clustering with minimal spanning tree. Model based Clustering: Expectation Maximization Algorithm. Density Based Clustering: DBSCAN, Evaluating clustering tendency, Evaluation of clusters	6	20	25	20	PI 1.4.2 PI 2.4.3 PI 4.3.5
Learning Outcome-4.1	Explain different clustering algorithms including graph based, model-based, and density-based methods. (PI 1.4.2) (CO2)					
Learning Outcome-4.2	Apply clustering algorithms to group unlabelled data. (PI 2.4.3) (CO4)					
Learning Outcome-4.3	Evaluate the quality and tendency of clusters using appropriate metrics. (PI 4.3.5) (CO4)					
5	Introduction to Neural Networks Biological neuron, models of a neuron, Introduction to Neural networks, network architectures (feedforward, feedback etc.), Perceptron, training a Perceptron,	4	12	10	20	PI 1.4.1 PI 1.4.3 PI 2.1.5

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
	Multilayer Perceptron, Limitations of MLP.					
Learning Outcome-5.1	Recall the structure and function of biological and artificial neurons. (PI 1.4.1) (CO1)					
Learning Outcome-5.2	Implement simple neural networks including Perceptron and Multilayer Perceptron. (PI 1.4.3) (CO3)					
Learning Outcome-5.3	Identify the limitations of neural networks and propose improvements. (PI 2.1.5) (CO5)					
6	Introduction to optimization Introduction to optimization in ML, Role of Loss Functions and Optimization. Case Study.	2	8	0	20	PI 1.4.2 PI 2.4.4 PI 3.4.6
Learning Outcome-6.1	Explain the role of loss functions and optimization in machine learning. (PI 1.4.2) (CO4)					
Learning Outcome-6.2	Analyze different optimization strategies used in model training. (PI 2.4.4) (CO5)					
Learning Outcome-6.3	Design an optimized machine learning solution based on a given case study. (PI 3.4.6) (CO5), (CO6)					
	* Insert rows for more modules in the Course **Total**	**30**	**100**	**100**	**100**	

Note: - As an attachment Annexure is required for assessment criteria of learning outcomes.

2.b Prerequisite Courses

No.	Semester	Name of the Course	Topic/s
1	4	Python Skill Based Programming	Python Data Structures, NumPy and Pandas Basics, Matplotlib Basics
2	4	Engineering Mathematics IV	Matrix Operations (Linear Algebra), Probability & Statistics, Differentiation (Calculus)
3	5	Data Warehousing and Mining	Data Preprocessing, Classification and Clustering, Association Rule Mining

2.c Relevance to Future Courses

No.	Semester	Name of the Course
1	VII	AI DS, Optimization in Machine Learning
2	VII	Natural Language Processing
3	VIII	Big Data Analytics

2.d See :- Identify real life scenarios/examples which uses the knowledge of the subject ,(Discussion on how to prepare examples and case studies e.g. "Boeing Plane": C Programming Language – Intro to Computer Science – Harvard's CS50 (2018) – Bing video)

Real Life Scenario	Concept Used
Recommendation Systems, Prediction System	Mathematical concepts used in Machine Learning algorithms
Sentiment Analysis, Medical Diagnosis	
Image and Speech Recognition	

3 Past Results – Division-Wise

Details	Target – DEC 2026	MAY 2025	DEC 2024	MAY 2024
Course Passing % – Average of 2 Divisions	100	97.45%	--	
Marks Obtained by Course Topper (mark/75)	75	71	-	-

	Division A		Division B		Division C	
Year	Initials of Teacher	% Result	Initials of Teacher	% Result	Initials of Teacher	% Result
May 2025	SDG	96.52	SM	97.31	SDG	95.54
May 2024	-	-	-	--	-	-
May 2023	-	-	-	-	-	-

4 All the Learning Resources – Books and E-Resources

4.a List of Textbooks (T – Symbol for Textbooks) to be Referred by Students

Sr. No	Textbook Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Introduction to Machine Learning	Ethem Alpaydın	MIT Press	4th	2,3,4,5	Yes
2	Machine Learning in Action	Peter Harrington	Manning Publication	1st	1,2,3,4	Yes
3	Deep Learning	Ian Goodfellow, Yoshua Bengio, Aaron Courville	MIT Press	1st	2,3,4,5	No
4	Machine Learning	Tom M. Mitchell	McGraw Hill	1st	2,3,4	Yes

4.b List of Reference Books (R – Symbol for Reference Books) to be Referred by Students

Sr. No	Reference Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Machine Learning for beginners	Harsh Bhasin	BPB Publication	-	All	Yes

4.c List of E - Books (E – Symbol for E-Books) to be Referred by Students

Sr. No	E- Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Understanding Machine Learning: From Theory to Algorithms	Shai Shalev-Shwartz and Shai Ben-David	Cambridge University Press	2014	1,2,3,4,5	No
2	An Introduction to Optimization	Edwin Chong	Wiley	2001	6	No

4.d Reading latest / top rated research papers (at least 5 papers)

Name of Paper	Name of Authors (Background)	Published in		Problem Statement	Available in our Library
		Date	Journal		
A Survey on Learning from Data with Label Noise via Deep Neural Networks	Baoye Song, Shihao Zhao, Luyao Dang, Haoguang Wang, Lin Xu	04 April 2025	Information Security Journal (Taylor & Francis)	Machine learning models often suffer when training data contains incorrect or noisy labels, leading to poor prediction accuracy.	No
A_Survey_on_Deep_Active_Learning_Recent_Advances_and_New_Frontiers	Dongyuan Li et. Al.	April 2025	IEEE Transactions on Neural Networks and Learning Systems	Traditional clustering methods struggle with high-dimensional and complex datasets.	No
A Comprehensive Survey of Deep Learning for Time Series Forecasting: Architectural Diversity and Open Challenges	Jongseon Kim, Hyungjoon Kim, HyunGi Kim,	23 April 2025	Artificial Intelligence Review (Springer)	Accurate forecasting of time-series data remains challenging	No

	Dongjun Lee, Sungroh Yoon et al.			because of nonlinear patterns and long-term dependencies.	
Deep Learning for Time Series Forecasting: A Survey	Xiangjie Kong, Zhenghao Chen, Weiyao Liu, Kaili Ning, Feng Xia et al.	08 February 2025	International Journal of Machine Learning and Cybernetics (Springer)	Selecting suitable deep learning architectures for forecasting remains difficult due to rapidly evolving models.	No
Machine learning based screening of potential paper mill publications in cancer research	Baptiste Scancar et. Al.	January 2026	Biomedical Journal	To train and validate a machine learning model to distinguish paper mill publications from genuine cancer research articles, and to screen the cancer research literature to assess the prevalence of papers that have textual similarities to paper mill papers.	No

4.e **Based on research paper an identify the current Problem statement**

Problem Statement			Used in			
	Quiz	Assignment	*Mini Project*	Poster Presentation	Test	Any Other
Comparing various machine learning	✓	✓	✓	✓		

algorithms for a specific applications						

4.f Identify Companies / Industries which use the knowledge of the subject and thus may provide Internships and final Placements

Name of the Company	To be / Contacted for		
	Student Internship	Student Final Placement	Faculty Internship
DesSecurity	✓		
Capgemini	✓		
TCS	✓		
C-DAC	✓		

4.g Identify suitable relevant TOP Guest Speakers from Industry, Example: - (CS50 Lecture by Mark Zuckerberg - 7 December 2005 - YouTube)

Name of the Identified Guest Speaker	Designation	Name of the Company
Mr. Laxmikant Tiwari	Associate VP	IGP.COM
Mr. Tushar Kamble	NLP Engineer	Ascendion

4.h Identify relevant technical competitions to participate [Competitions -Paper Presentations, Projects, Hackathons, IVs etc..]

Name of the Relevant Technical Competition Identified to participate	Organized by	Date of the Event
Smart India Hackathon 2026	AICTE	August–December 2026
Titanic - Machine Learning from Disaster	Kaggle	July 2026
House Prices - Advanced Regression Techniques	Kaggle	July 2026
Kaggriculture	Kaggle	Sept 2026
TIH crop- disease detection	IITB	July 2026

4.i Identify faculty in TOP schools / Universities who are teaching same / similar subject and develop rapport e.g. Exchange Lecture Material (Assignments / Tests / Project etc..), Joint Paper Publication

University	Name of the Course	Name of Faculty	Type of Collaboration		
			Exchange of Lecture Material	Joint Publication/ Research	Other
MITOCW	Introduction to Machine Learning	Prof. Leslie Kaelbling	Introduction to Machine Learning | Electrical Engineering and Computer Science | MIT OpenCourseWare	-	-

	MIT Course Number 6.036				
Stanford CS229	Machine Learning	Andrew Ng	https://www.youtube.com/watch?v=jGwO_UgTS7I&list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU	-	Assignment

4.j **Module Best Available in – Title of the best resource [from *4.a* to *4.d* in this AAP] and other details as necessary**

Module No.	Title of the Module	Textbook	Mention the Tile					
			Reference Book	E-books	Journal	E-Journal	Available in our Library	V-refer
1	Introduction to Machine Learning		✓					
2	Supervised Learning-I		✓					
3	Supervised Learning-II					✓		
4	Unsupervised Learning					✓		
5	Introduction to Neural Networks					✓		
6	Introduction to optimization					✓		

4.k Referred to any top-rated university in that subject for content

University	Name of the Course	Name of Faculty	Date of Delivery of the Course	Remarks
Massachusetts Institute of Technology, (MIT)	Machine Learning MIT Course Number 6.867	Prof. Tommi Jaakkola	2016	This course introduces principles, algorithms, and applications of machine learning from the point of view of modeling and prediction. It includes formulation of learning problems and concepts of representation, over-fitting, and generalization. 6.867 Machine Learning (mit.edu)
Standford university	Various course related to machine learning and advance	Prof. Andrew Ag	2018	https://www.linkedin.com/posts/analyticalrohit_top-10-stanfords-courses-on-ai-ml-if-share-7475172864447664128-hadg?utm_source=social_share_send&utm_medium=android_app&rcm=ACoAAESn9hoBEm5cA7vOYmgSH7XvbDpFlkWBVUs&utm_campaign=whatsapp

4.l Faculty received any certification related to this subject. List of Certifications Identified / Done

Course	Certifying Agency	No. of Hours	Level of the Course		Certification		Remarks
			Introductory	Advance Skill Development	Done on	Proposed to be on	
Introduction to machine Learning	NPTEL	12 Weeks	▪		April 2023		
Business Statistics	NPTEL	12 Weeks	▪		April 2023		

4.m Completed subject wise/cluster wise training with cluster mentor.

List of relevant Refresher Course Identified / Done

Course	Certifying Agency (As suggested by DAB/Cluster Mentor/Industry/University other than MU)	Certification		Remarks
		Done on	Proposed to be on	
Pedagogy	Coursive	11 July 2026		MASTER OF CHATGPT
PBL				
Sub. Content Training	Introduction to Machine Learning		In progress	October 18, 2026

4.n Best Practices Identified and adopted

No.	Item	Best Practices Identified		
		Massachusetts Institute of Technology (MIT)	Standford University	University of Cambridge
1	Microsite		https://lnkd.in/eZPepJrM	
2	Video Lectures	✓	✓	✓
3	Assignments	✓	✓	✓
4	Mini Project			
5	Assessment Metric			
6	Quizzes	✓		✓
7	Labs/ Practical (PBL)			
8	Tests			
9	Peer Assessment			
10	Any Other			

4.o Web Links for Online Notes/YouTube/VIT Digital Content/VIT Lecture Capture/NPTEL Videos

Students can view lectures by VIT professors, captured through LMS 'Lecture Capture' in VIT campus for previous years.

No.	Websites / Links	Module Nos.
1	https://www.tutorialspoint.com/machine_learning_with_python/index.htm	All
2	https://www.youtube.com/watch?v=PPLop4L2eGk&list=PLLssT5z_DsK-h9vYZkQkYNWcItqhlRJLN	All

4.p Recommended MOOC Courses like Coursera / NPTEL / MIT-OCW / edX/VAC etc.

Sr. No.	MOOC Course Link	Course conducted by – Person / University / Institute / Industry	Course Duration	Certificate (Y / N)
1	https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs133	By Prof. Balaraman Ravindran	12 weeks	Y
2	Machine Learning using Python Course Link: https://www.simplilearn.com/learn-machine-learning-basics-skillup	Simplilearn SkillUp	Self pace	Y

5 Consolidated Course Lesson Plan

	From (date/month/year)	From (date/month/year)	Total Number of Weeks
Semester Duration	6 July 2026	16 October 2026	15

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
1	1	1	Introduction to Machine Learning Techniques: Supervised Learning, Unsupervised Learning and Reinforcement Learning		CO1	BL1		1/T1/2
	2	1	Data formats, Applications, Feature Selection and Filtering		CO1	BL2		3/T1/74
2	3	1	Dimensionality Reduction Techniques		CO1	BL3		1/T2/39
	4	1	Principal Component Analysis		CO1	BL3		6/T2/245
3	5	1	Linear Discriminant Analysis		CO1	BL3		6/T2/245

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
	6	1	Singular Valued Decomposition		CO1	BL3		23/R2/97 0
4	7	2	Regression: Linear regression models, Nonlinear regression (only introduction)		CO2	BL3		23/R2/97 0
	8	2	SVM classifier Support Vector Machine classification algorithm		CO2	BL3		23/R2/97 0
5	9	2	hyper plane, optimal separating hyperplanes		CO2	BL4		10/E1/13 0
	10	2	kernel functions, kernel selection, applications		CO2	BL4		10/E1/13 0
6	11	2	Introduction to random forest, growing of random forest, random feature selection		CO2	BL2		10/E1/ 13 0
	12	2	Introduction to random forest, growing of random forest, random feature selection		CO2	BL2		15/E1/20 9

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
7	13	3	Evaluation of classifiers: Accuracy, Precision, Recall, F1 score		CO3	BL3		15/E1/20 9
	14	3	TPR, TNR, Confusion matrix		CO3	BL3		15/E1/20 9
8	15	3	ROC, Overfitting, Underfitting		CO3	BL3		22/E1/31 5
	16	3	Variance, Bias, Concepts of regularization and generalization		CO3	BL3		22/E1/31 5
9	17	3	Ensemble Learning: Basic concept, Stacking, Bagging, Boosting		CO3	BL3		22/E1/31 5
	18	3	Random Forest, AdaBoost, GBM, XG Boost		CO3	BL3		
10			Non-Instructional Week					
			Non-Instructional Week					
11	19	4	Types of Clustering algorithms, Graph Based		CO4	BL2		23/R2/97 0

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
			Clustering: Clustering with minimal spanning tree					
	20	4	Types of Clustering algorithms, Graph Based Clustering: Clustering with minimal spanning tree		CO4	BL2		23/R2/970
12	21	4	Model based Clustering: Expectation Maximization Algorithm		CO4	BL3		10/E1/130
	22	4	Density Based Clustering: DBSCAN		CO4	BL3		10/E1/130
13	23	4	Evaluating clustering tendency		CO4	BL3		10/E1/130
	24	4	Evaluation of clusters		CO4	BL4		15/E1/209
14	25	5	Biological neuron, models of a neuron, Introduction to Neural networks		CO5	BL3		15/E1/209
	26	5	network architectures (feedforward, feedback etc.),		CO5	BL3		15/E1/209

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
15	27	5	network architectures (feedforward, feedback etc.),		CO5	BL3		22/E1/31 5
	28	5	Perceptron, training a Perceptron, Multilayer Perceptron, Limitations of MLP		CO5	BL3		22/E1/31 5
16	29	2	Introduction to optimization in ML, Role of Loss Functions and Optimization		CO5	BL3		23/R2/ 97 0
	30	2	Case Study		CO5	BL5		10/E1/ 13 0

6 Rubric for Grading and Marking of Term Work (inform students at the beginning of semester)

- Activity/ies should be designed as per reference of credit structure.
- If the subject is of 2 credit, activity/ assignment should be design for 2 hours with appropriate complexity and engaging time.

Theory (ISA=15)	Total

Class Participation	Assignment -1	Assignment -2	Assignment -3	Assignment -4	Assignment -5	Assignment -6	Assignment -7	Assignment -8	Assignment -9	Assignment -10	Assignment -11	
	15M	15M	15M	15M	15M	15M	15M	15M	15M	15M	15M	15M

Practical (ISA=____)												Total
Class Participation	Practical -1	Practical -2	Practical -3	Practical -4	Practical -5	Practical -6	Practical -7	Practical -8	Mini Project			
5M	10M	10M	10M	10M	10M	10M	10M	10M	10M			25M

Class Participation	MSE-1	MSE-2	ESE	Total
15 M	20M	20M	40M	75M

7 Assignments / Tutorials Details

Assignment/ Tutorial No.	Title of the Assignments / Tutorials	CO Map	Mapping Bloom Taxonomy Level	Assignment/ Tutorials given to Students on	Assignments to be submitted back on
1	Introduction to Machine Learning: AI vs ML vs DL, Types of ML, Applications and ML Workflow	CO1	BL2	WEEK1	WEEK2
2	Data Preprocessing and Feature Engineering using Python (Handling Missing Values, Encoding, Scaling, Feature Selection)	CO2	BL3	WEEK2	WEEK3
3	Linear and Logistic Regression: Implementation and Comparative Analysis using Real-world Dataset	CO3	BL3	WEEK3	WEEK4
4	Classification Algorithms: Decision Tree, k-NN and Naïve Bayes – Performance Comparison	CO3	BL4	WEEK4	WEEK5
5	Model Evaluation Techniques: Confusion Matrix, Precision, Recall, F1-Score, ROC Curve and Cross Validation	CO4	BL4	WEEK5	WEEK6
6	Ensemble Learning Methods: Bagging, Random Forest, Boosting and Stacking	CO4	BL4	WEEK6	WEEK7
7	Unsupervised Learning: Clustering using K-Means, Hierarchical Clustering and DBSCAN	CO4	BL3	WEEK7	WEEK8
8	Artificial Neural Networks: Perceptron, Multilayer Perceptron and Deep Learning Basics	CO5	BL5	WEEK8	WEEK9
9	Hyperparameter Optimization and Model Selection using Grid Search and Random Search	CO5	BL5	WEEK9	WEEK10
10	Case Study: Design and Development of a Complete Machine Learning Solution for a Real-world Application (Healthcare, Agriculture, Finance, Smart City, etc.)	CO6	BL6	WEEK10	WEEK11
11	Poster presentation	ALL	BL2	WEEK11	WEEK12

Analysis of Assignment / Tutorial Questions and Related Resources

Assignment / Tutorial No.	Week No.	Type* (√)			Module No.	Based on #			Question Type (√)	
		OT	CS	DTP		Textbook	Reference Book	Other Learning Resource	Real Life Assignments	Thought Provoking
1	3rd	√	-	-	1,2,3	T1,T2, R2	-	E1	-	√
2	4th	√	-	-	1,2,3	T4, T5	-	E1	-	√
3	5th	√	-	-	2	T1,T2, R2	-	E1	√	-
4	6th	√	-	-	2	T4, T5	-	E1	√	-
5	7th	-	√	-	3	T1,T2, R2	-	E1	-	√
6	8th	√	-	-	3	T4, T5	-	E1	-	√
7	9th	√	-	-	All	T1,T2, R2	-	E1	-	√
8	10th	√	-	-	4	T4, T5	-	E1	-	√
9	12th	√	-	-	5	T1,T2, R2	-	E1	-	√
10	13th	√	-	-	ALL	T4, T5	-	E1	-	√
11	14th	√	-	-	ALL	T1,T2, R2	-	E1	-	√

* Tick (√) the Type of the Assignment: Online Tools (OT); Collaborative Assignments (CS); Design /Thought provoking (DTP)
Write number for textbook, reference book, other learning resource from this AAP – *from Points 4.a to 4.d*

8 In Semester Assessment (ISE) / Other Class Test / Open Book Test (OBT)/Take Home Test (THT) Details

Tests	Test Dates	Module No.	CO Map	MSE Question Paper Pattern	Policy
MSE1	10/8/2026	1,2	CO1, CO2	Q1 – Numerical and theory Questions - 10 Marks Q2 – 2 Numerical Questions 10 Marks	MSE is a Head of passing *
MSE2	28/9/2026			Q1 – Numerical and theory Questions - 10 Marks Q2 – 2 Numerical Questions 10 Marks	
Pop Quiz	7th Week	1	CO1	Objective type of questions	
Open Book Test	Week3	1,2	CO1, CO2		
Take Home Test	Week9	5	CO5		

Class tests / prelims					
Class tests / prelims					
Any other test/exams					

*** Failures of IA test (IA1+IA2) shall appear for IA test in the next semester. There is no provision for re-test in the same semester.**

9. Practical Activities

Practical No.	Module No.	Title of the **Experiments**	Type of Experiment		Topics to be highlighted	CO Map
			PBL	Newly Added		
1	M1	Cleaning the Hospital Data for Prediction	Yes		Understand the importance of preprocessing in real-world ML tasks	CO3, CO5
2	M2	Predicting Customer Churn with Logistic Regression			Demonstrate logistic regression and evaluate its predictions	CO3, CO4
3	M2, M3	Forest Classifier in the Wild			Compare model performance with different hyperparameter settings	CO3, CO5
4	M1	Separating Emotions in 2D using LDA	Yes		Understand supervised dimensionality reduction with LDA	CO3, CO5
5	M3	Boosting vs Gradient Boosting for Spam Detection			Evaluate and compare ensemble learning methods	CO4, CO5
6	M3	Battle of the Classifiers – ROC Challenge			Visually analyze and interpret model performance	CO4
7	M4	Is There Any Cluster at All?	Yes		Understand unsupervised learning and cluster evaluation	CO2, CO3, CO4
8	M5	Training a Tiny Brain – MLP			Understand the basics of neural networks and training challenges	CO3, CO4, CO5
9	M1	When PCA Meets Image Compression			Demonstrate dimensionality reduction in real-world data	CO3, CO5
10	M6	Optimization in Action: Comparing Loss Functions			Analyze role of optimization in training performance.	CO4, CO5

10 Uncovering syllabus with different Activities.

No.	Type of the Activity	Activities	Number of beneficiaries	Other Details – guest profile, feedback, mark sheet, report
1	**Experiential learning/Interaction with Outside World**	1- Guest Lectures by Industry Expert	226	Week 10
		2- Workshops		
		3- Mini Project	226	Week 12
		4- Industrial Visit	34	25th July 2026
		5- Any other activity		
2	**Collaborative & Group Activity**	6- Poster Presentation	226	Week 11
		7- Minute Papers		
		8- Students Seminars		
		9- Students Debates		
		10- Panel Discussion / Mock GD		
		11- Mock Interview		
		12- Any other activity		
3	**Co-Curricular Activity**	13- Informative videos (NPTEL/YouTube /TEDx/ MIT OW/edX)	226	Week 7
		14- Lecture Capture Usage		
		15- Any other activity		
4	**Tests & Assessments**	16- Class Tests/ Weekly Tests		
		17- Pop Quiz	226	Week 5
		18- Mobile App Based Quiz		
		19- Open Book Test		

		20- Take Home Test		
		21- Any other activity		

11 AAP

No.	Programme	Course	Uploaded on V-refer	Date
1	INFT	Machine learning	Yes	Aug 2026

12 Lecture Guide

No.	Programme	Course	Uploaded on V-refer	Date
1	INFT	Machine learning	Yes	Aug 2026

*** Do not delete any activity. Give details for planned events. Write 'NA' for activity Not Planned.**

Consolidated Academic Administration Plan Prepared by (mention all theory teaching faculty names with signature)

Please write below your name and sign with date of the external cluster mentor meeting

Kanchan Dhuri Faculty 1	Faculty 2	Faculty 3

External Industry Mentor	External Academic Mentor	VIT Cluster Mentor	Program HOD

Annexure:

Assessment Criteria of Learning Outcomes:

Learning Outcomes: The Learner will:	Assessment Criteria: The Learner can:	Evaluated under ISA/MSE/ESE/LAB
LO1: Explain the fundamental concepts of Machine Learning, its learning paradigms, applications, and dimensionality reduction techniques.	1.1 Explain the concepts of supervised, unsupervised, and reinforcement learning with suitable examples.	ISA, MSE, ESE
	1.2 Differentiate between feature selection and feature extraction, and explain the need for dimensionality reduction techniques such as PCA and LDA.	ISA, MSE, ESE
	1.3 Identify suitable Machine Learning techniques for real-world applications and justify their selection.	ISA, ESE
LO2: Apply regression, Support Vector Machine (SVM), and Random Forest algorithms to solve supervised learning problems.	2.1 Explain the working principles of linear and nonlinear regression models.	ISA, MSE
	2.2 Apply Linear Regression and SVM algorithms to solve prediction and classification problems.	MSE, ESE
	2.3 Compare the performance and applications of SVM and Random Forest for different supervised learning tasks.	ISA, ESE
LO3: Analyze the performance of machine learning models using evaluation metrics, regularization, and ensemble learning techniques.	3.1 Compute and interpret classification performance metrics such as Accuracy, Precision, Recall, F1-score, and ROC.	ISA, MSE
	3.2 Analyze the impact of overfitting, underfitting, bias, variance, and regularization on model performance.	MSE, ESE
	3.3 Compare ensemble learning techniques such as Bagging, Boosting, Stacking, and Random Forest for improving prediction accuracy.	ISA, ESE
LO4: Apply clustering algorithms and evaluate clustering quality for unlabeled datasets.	4.1 Explain the working principles of graph-based, model-based, and density-based clustering algorithms.	ISA, MSE
	4.2 Apply appropriate clustering algorithms to group unlabeled data based on similarity.	MSE, ESE
	4.3 Evaluate clustering quality using suitable evaluation metrics and justify the selected clustering method.	ISA, ESE
LO5: Explain the structure of neural networks and apply Perceptron and Multilayer Perceptron models to basic machine learning problems.	5.1 Explain the structure and working of biological neurons, artificial neurons, and neural network architectures.	ISA, MSE
	5.2 Apply Perceptron and Multilayer Perceptron models for solving simple classification problems.	MSE, ESE

	5.3 Analyze the limitations of Perceptron and Multilayer Perceptron and suggest suitable improvements.	ISA, ESE
LO6: Explain the role of loss functions and optimization in machine learning.	6.1 Define optimization and explain its significance in machine learning model training.	ISA, MSE, ESE
	6.2 Explain different types of loss functions (e.g., Mean Squared Error, Cross-Entropy Loss) and their applications.	ISA, MSE, ESE
	6.3 Describe how optimization algorithms minimize the loss function to improve model performance.	MSE, ESE